



Microsoft Sentinel Home Lab with Azure Bicep

This guide deploys a small Microsoft Sentinel lab using **Bicep**. The lab is designed for learning Infrastructure as Code (IaC), Azure Activity Logs, Kusto Query Language (KQL), Sentinel analytics rules, incident generation, and cleanup.

The initial version does not deploy a virtual machine, Defender plans, a Logic App, or other continuously running resources. This keeps the lab inexpensive while still producing useful security events.

Cost warning: Microsoft Sentinel and Log Analytics can incur charges based on data ingestion and retention. Pricing, free allowances, and supported features vary by region and subscription type. Check current Azure pricing before deploying, create a budget alert, and delete the lab when finished.

1. What You Will Build

```
Azure subscription
|
|-- Resource group: rg-sentinel-homelab
|   |-- Log Analytics workspace
|   |-- Microsoft Sentinel
|   |-- Scheduled analytics rules
|
|-- Subscription Activity Log diagnostic setting
|   |-- Sends administrative, security, policy, and health events to the workspace
```

You will test the following flow:

```
Azure resource change
    |
    v
Subscription Activity Log
```

```

    |
    v
Log Analytics: AzureActivity table
    |
    v
Sentinel scheduled analytics rule
    |
    v
Alert and incident

```

2. Learning Objectives

By the end of the lab, you should be able to:

- Deploy Azure resources from a Bicep template.
- Validate and redeploy Bicep safely.
- Enable Microsoft Sentinel on a Log Analytics workspace.
- Send subscription Activity Logs to Log Analytics.
- Write and test KQL queries against `AzureActivity`.
- Create Sentinel scheduled analytics rules as code.
- Generate a controlled test event.
- Find the resulting alert and incident.
- Remove the lab without leaving subscription-level configuration behind.

3. Prerequisites

Azure requirements

- An Azure subscription where you are allowed to create resources.
- Permission to create a resource group, workspace, diagnostic setting, and Sentinel resources.
- Permission to create or remove a temporary resource group for testing.
- A browser for the Azure portal.

For the simplest setup, use an account with **Owner** or **Contributor** access plus the required Log Analytics and Microsoft Sentinel permissions. In a managed organization, ask an administrator to grant only the roles required by your organization's policy.

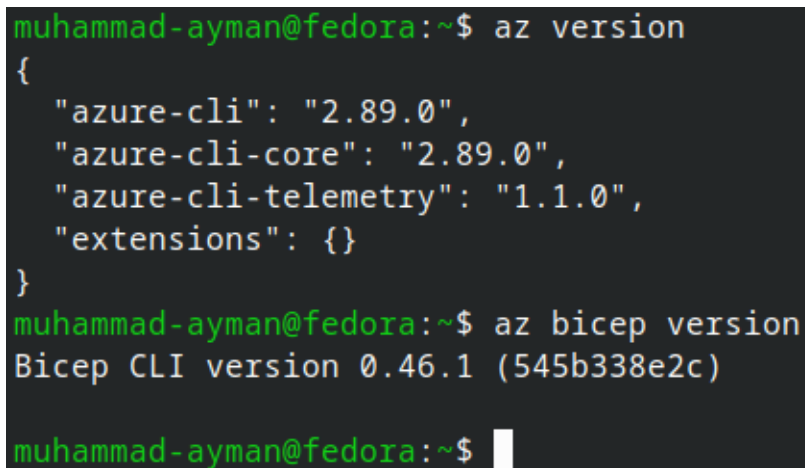
Local tools

Install:

- [Azure CLI](#)
- [Bicep CLI](#)
- A text editor, such as Visual Studio Code with the Bicep extension

Verify the tools:

```
az version
az bicep version
```



```
muhammad-ayman@fedora:~$ az version
{
  "azure-cli": "2.89.0",
  "azure-cli-core": "2.89.0",
  "azure-cli-telemetry": "1.1.0",
  "extensions": {}
}
muhammad-ayman@fedora:~$ az bicep version
Bicep CLI version 0.46.1 (545b338e2c)
muhammad-ayman@fedora:~$
```

4. Create the Lab Directory

Create a working directory and move into it:

```
mkdir sentinel-bicep-homelab
cd sentinel-bicep-homelab
```

Create a file named `main.bicep` in this directory and paste in the template from the next section.

```
muhammad-ayman@fedora:~/Downloads$ mkdir sentinel-bicep-homelab
muhammad-ayman@fedora:~/Downloads$ cd sentinel-bicep-homelab
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ touch main.bicep
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$
```

5. Create the Bicep Template

The deployment uses two scopes. `main.bicep` deploys at subscription scope. `modules/sentinel.bicep` deploys at resource-group scope and creates the Log Analytics workspace, Sentinel onboarding, and two scheduled analytics rules.

Save this as `main.bicep`:

```
targetScope = 'subscription'

@description('Azure region for the resource group and workspace.')
param location string = 'eastus'

@description('Dedicated resource group for this lab.')
param resourceGroupName string = 'rg-sentinel-homelab'

@description('Globally unique Log Analytics workspace name.')
param workspaceName string = 'law-sentinel-homelab-${uniqueString(subscription().id)}'

@description('Workspace retention in days. Increase only when needed.')
@minValue(30)
param retentionInDays int = 30

@description('Name of the subscription Activity Log diagnostic setting.')
param diagnosticSettingName string = 'send-activity-to-sentinel'

resource labResourceGroup 'Microsoft.Resources/resourceGroups@2023-07-01' = {
  name: resourceGroupName
  location: location
  tags: {
    environment: 'homelab'
    managedBy: 'bicep'
    purpose: 'microsoft-sentinel-learning'
  }
}

module sentinelLab 'modules/sentinel.bicep' = {
  name: 'sentinel-lab-resources'
  scope: labResourceGroup
}
```

```

params: {
  location: location
  workspaceName: workspaceName
  retentionInDays: retentionInDays
}
}

resource activityDiagnosticSetting 'Microsoft.Insights/diagnosticSettings@2021-05-01-preview' = {
  scope: subscription()
  name: diagnosticSettingName
  properties: {
    workspaceId: sentinelLab.outputs.workspaceResourceId
    logs: [
      {
        category: 'Administrative'
        enabled: true
      }
      {
        category: 'Security'
        enabled: true
      }
      {
        category: 'Policy'
        enabled: true
      }
      {
        category: 'ServiceHealth'
        enabled: true
      }
      {
        category: 'ResourceHealth'
        enabled: true
      }
    ]
  }
}

output resourceId string = labResourceGroup.id
output workspaceResourceId string = sentinelLab.outputs.workspaceResourceId
output workspaceName string = sentinelLab.outputs.workspaceName

```

```

muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ nano main.bicep

```

```
GNU nano 8.7.1 main.bicep
targetScope = 'subscription'

@description('Azure region for the resource group and workspace.')
param location string = 'eastus'

@description('Dedicated resource group for this lab.')
param resourceGroupName string = 'rg-sentinel-homelab'

@description('Globally unique Log Analytics workspace name.')
param workspaceName string = 'law-sentinel-homelab-${uniqueString(subscription().id)}'

@description('Workspace retention in days. Increase only when needed.')
@minValue(30)
param retentionInDays int = 30

@description('Name of the subscription Activity Log diagnostic setting.')
param diagnosticSettingName string = 'send-activity-to-sentinel'

resource labResourceGroup 'Microsoft.Resources/resourceGroups@2023-07-01' = {
  name: resourceGroupName
  location: location
  tags: {
    environment: 'homelab'
    managedBy: 'bicep'
    purpose: 'microsoft-sentinel-learning'
  }
}

resource workspace 'Microsoft.OperationalInsights/workspaces@2023-09-01' = {
  scope: labResourceGroup
  name: workspaceName
  location: location
  tags: {
    environment: 'homelab'
    managedBy: 'bicep'
    purpose: 'microsoft-sentinel-learning'
  }
  properties: {
    sku: {
      name: 'PerGB2018'
    }
  }
}
```

Create the module directory and save this as `modules/sentinel.bicep`:

```
targetScope = 'resourceGroup'

@description('Azure region for the workspace.')
param location string

@description('Globally unique Log Analytics workspace name.')
param workspaceName string

@description('Workspace retention in days.')
@minValue(30)
param retentionInDays int = 30

var labTags = {
  environment: 'homelab'
  managedBy: 'bicep'
  purpose: 'microsoft-sentinel-learning'
}

resource workspace 'Microsoft.OperationalInsights/workspaces@2023-09-01' = {
  name: workspaceName
  location: location
  tags: labTags
  properties: {
    sku: {
      name: 'PerGB2018'
    }
    retentionInDays: retentionInDays
    features: {
```

```

        enableLogAccessUsingOnlyResourcePermissions: true
      }
    }
  }

resource sentinel 'Microsoft.SecurityInsights/onboardingStates@2023-02-01' = {
  name: 'default'
  scope: workspace
  properties: {}
}

resource networkChangeRule 'Microsoft.SecurityInsights/alertRules@2023-02-01' = {
  name: guid(subscription().id, 'sentinel-network-change-rule')
  scope: workspace
  dependsOn: [
    sentinel
  ]
  kind: 'Scheduled'
  properties: {
    displayName: 'Lab - Network security group changed'
    description: 'Detects creation, update, or deletion of a network security group.'
    severity: 'Low'
    enabled: true
    query: '''
      AzureActivity
      | where CategoryValue == "Administrative"
      | where ResourceProviderValue =~ "Microsoft.Network"
      | where OperationNameValue has_any ("networkSecurityGroups", "securityGroups")
      | where ActivityStatusValue == "Success"
      | project TimeGenerated, Caller, OperationNameValue, ResourceGroup, ResourceId, Act
ivityStatusValue
      '''
    queryFrequency: 'PT5M'
    queryPeriod: 'PT10M'
    triggerOperator: 'GreaterThan'
    triggerThreshold: 0
    suppressionEnabled: false
    suppressionDuration: 'PT5H'
    incidentConfiguration: {
      createIncident: true
      groupingConfiguration: {
        enabled: false
        lookbackDuration: 'PT5H'
        matchingMethod: 'AllEntities'
        reopenClosedIncident: false
      }
    }
  }
  tactics: [
    'DefenseEvasion'
  ]
}
}

```

```

resource policyChangeRule 'Microsoft.SecurityInsights/alertRules@2023-02-01' = {
  name: guid(subscription().id, 'sentinel-policy-change-rule')
  scope: workspace
  dependsOn: [
    sentinel
  ]
  kind: 'Scheduled'
  properties: {
    displayName: 'Lab - Azure Policy assignment changed'
    description: 'Detects a successful Azure Policy assignment write or delete operation.'
    severity: 'Medium'
    enabled: true
    query: ''
      AzureActivity
      | where CategoryValue == "Administrative"
      | where OperationNameValue has "policyAssignments"
      | where ActivityStatusValue == "Success"
      | project TimeGenerated, Caller, OperationNameValue, ResourceGroup, ResourceId, ActivityStatusValue
    ''
    queryFrequency: 'PT5M'
    queryPeriod: 'PT10M'
    triggerOperator: 'GreaterThan'
    triggerThreshold: 0
    suppressionEnabled: false
    suppressionDuration: 'PT5H'
    incidentConfiguration: {
      createIncident: true
      groupingConfiguration: {
        enabled: false
        lookbackDuration: 'PT5H'
        matchingMethod: 'AllEntities'
        reopenClosedIncident: false
      }
    }
    tactics: [
      'DefenseEvasion'
    ]
  }
}

output workspaceResourceId string = workspace.id
output workspaceName string = workspace.name

```

Notes about the template

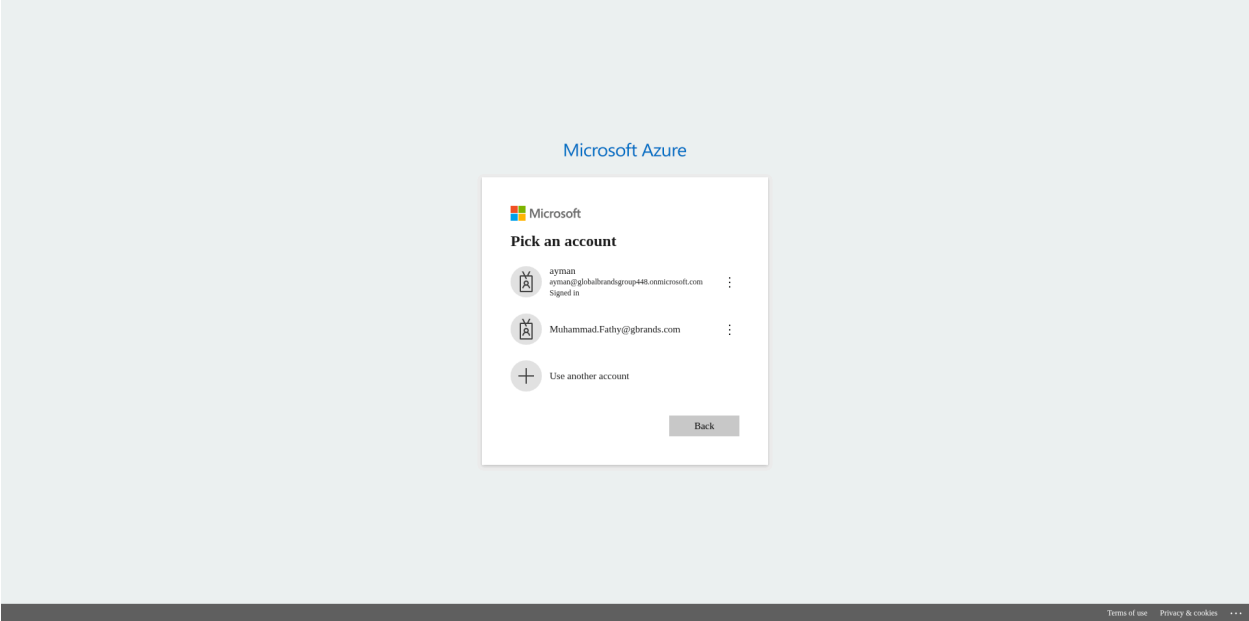
- `targetScope = 'subscription'` is required because the diagnostic setting is attached to the subscription.
- The module is required because Bicep does not allow a subscription-scope file to directly deploy a resource-group-scoped workspace.
- Sentinel onboarding and analytics rules use `scope: workspace`; Microsoft Sentinel is associated with a Log Analytics workspace, not directly with the resource group.
- `uniqueString(subscription().id)` prevents a workspace-name collision when the default is used.
- The workspace retention is set to 30 days. This is a minimum-style lab setting, not a production recommendation.
- Activity Log ingestion and Sentinel behavior can take several minutes. Do not expect an incident immediately after an operation.
- The exact Activity Log operation names can differ by resource provider. The KQL queries intentionally use `has` and `has_any`; inspect actual events and refine them for your subscription.

6. Sign In and Select the Subscription

Sign in to Azure:

```
az login
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-home-lab$ az login
A web browser has been opened at https://login.microsoftonline.com/organizations/oauth2/v2.0/authorize. Please continue the login in the web browser. If no web browser is available or if the web browser fails to open, use device code flow with 'az login --use-device-code'.
```



You have logged into Microsoft Azure!
You can close this window, or we will redirect you to the [Azure CLI documentation](#) in 1 minute.

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az login
A web browser has been opened at https://login.microsoftonline.com/organizations/oauth2/v2.0/authorize. Please continue the login in the web browser. If no web browser is available or if the web browser fails to open, use device code flow with 'az login --use-device-code'.

Retrieving tenants and subscriptions for the selection...

[Tenant and subscription selection]

No      Subscription name      Subscription ID      Tenant
-----
[1] Ayman-Visual Studio Enterprise Su... global brands group
[2] * Lasheen- MPN global brands group
[3] Visual Studio Enterprise Subscrip... global brands group
[4] Visual Studio Enterprise Subscrip... global brands group
[5] Zeyad subscription - MPN global brands group

The default is marked with an *; the default tenant is 'global brands group' and subscription is 'Lasheen- MPN' (1132ce03-611b-4164-9042-413aeb38825b).

Select a subscription and tenant (Type a number or Enter for no changes): 1
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az login
A web browser has been opened at https://login.microsoftonline.com/organizations/oauth2/v2.0/authorize. Please continue the login in the web browser. If no web browser is available or if the web browser fails to open, use device code flow with 'az login --use-device-code'.

Retrieving tenants and subscriptions for the selection...

[Tenant and subscription selection]

No      Subscription name      Subscription ID      Tenant
-----
[1] Ayman-Visual Studio Enterprise Su... global brands group
[2] * Lasheen- MPN global brands group
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[4] Visual Studio Enterprise Subscrip... global brands group
[5] Zeyad subscription - MPN global brands group

The default is marked with an *; the default tenant is 'global brands group' and subscription is 'Lasheen- MPN' (1132ce03-611b-4164-9042-413aeb38825b).

Select a subscription and tenant (Type a number or Enter for no changes): 1

Tenant: global brands group
Subscription: Ayman-Visual Studio Enterprise Subscription - MPN (1132ce03-611b-4164-9042-413aeb38825b)

[Announcements]
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more about it and its configuration at https://go.microsoft.com/fwlink/?linkid=2271236

If you encounter any problem, please open an issue at https://aka.ms/azclibug

[Warning] The login output has been updated. Please be aware that it no longer displays the full list of available subscriptions by default.

muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$
```

Confirm the active subscription:

```
az account show --query '{name:name,id:id,tenantId:tenantId}'
--output table
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az account show --query '{name:name,id:id,tenantId:tenantId}' --output table
Name                                     TenantId
-----
Ayman-Visual Studio Enterprise Subscription - MPN  7b9d43-91d4-484d-b84d-1f0000000000
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$
```

Never run this lab against a production subscription unless the subscription owner has approved the deployment.

7. Validate the Bicep Template

Run the Bicep linter/build step:

```
az bicep build --file main.bicep
```

Validate the subscription deployment without creating resources:

```
az deployment sub validate \
  --name sentinel-homelab-validation \
  --location eastus \
  --template-file main.bicep \
  --parameters location=eastus
```

If validation fails, read the first reported resource and property error. Common causes are insufficient permissions, a region that does not support the selected service, or an API version mismatch in an older CLI installation.

```

muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az bicep build --file main.bicep
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az deployment sub validate \
  --name sentinel-homelab-validation \
  --location eastus \
  --template-file main.bicep \
  --parameters location=eastus
{
  "error": null,
  "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/providers/Microsoft.Resources/deployments/sentinel-homelab-validation",
  "location": "eastus",
  "name": "sentinel-homelab-validation",
  "properties": {
    "correlationId": "96daa23f-8273-4919-92d1-4e16e98d6c52",
    "debugSetting": null,
    "dependencies": [
      {
        "dependsOn": [
          {
            "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab/providers/Microsoft.Resources/deployments/sentinel-lab-resources",
            "resourceGroup": "rg-sentinel-homelab",
            "resourceName": "sentinel-lab-resources",
            "resourceType": "Microsoft.Resources/deployments"
          },
          {
            "apiVersion": "2025-04-01",
            "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab/providers/Microsoft.Resources/deployments/sentinel-lab-resources",
            "resourceGroup": "rg-sentinel-homelab",
            "resourceName": "sentinel-lab-resources",
            "resourceType": "Microsoft.Resources/deployments"
          }
        ],
        "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/providers/Microsoft.Insights/diagnosticSettings/send-activity-to-sentinel",
        "resourceName": "send-activity-to-sentinel",
        "resourceType": "Microsoft.Insights/diagnosticSettings"
      }
    ],
    "dependsOn": [
      {
        "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab",
        "resourceName": "rg-sentinel-homelab",
        "resourceType": "Microsoft.Resources/resourceGroups"
      }
    ],
    "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab/providers/Microsoft.Resources/deployments/sentinel-lab-resources",
  }
}

```

8. Deploy the Lab

Deploy the template:

```

az deployment sub create \
  --name sentinel-homelab \
  --location eastus \
  --template-file main.bicep \
  --parameters location=eastus

```

```

muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az deployment sub create \
> --name sentinel-homelab \
> --location eastus \
> --template-file main.bicep \
> --parameters location=eastus
{
  "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/providers/Microsoft.Resources/deployments/sentinel-homelab",
  "location": "eastus",
  "name": "sentinel-homelab",
  "properties": {
    "correlationId": "dc068712-a9e5-427f-a450-827afbe42bd1",
    "debugSetting": null,
    "dependencies": [
      {
        "dependsOn": [
          {
            "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab/providers/Microsoft.Resources/deployments/sentinel-lab-resources",
            "resourceGroup": "rg-sentinel-homelab",
            "resourceName": "sentinel-lab-resources",
            "resourceType": "Microsoft.Resources/deployments"
          },
          {
            "apiVersion": "2025-04-01",
            "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab/providers/Microsoft.Resources/deployments/sentinel-lab-resources",
            "resourceGroup": "rg-sentinel-homelab",
            "resourceName": "sentinel-lab-resources",
            "resourceType": "Microsoft.Resources/deployments"
          }
        ],
        "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/providers/Microsoft.Insights/diagnosticSettings/send-activity-to-sentinel",
        "resourceName": "send-activity-to-sentinel",
        "resourceType": "Microsoft.Insights/diagnosticSettings"
      }
    ],
    "dependsOn": [
      {
        "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab",
        "resourceName": "rg-sentinel-homelab",
        "resourceType": "Microsoft.Resources/resourceGroups"
      }
    ],
    "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab/providers/Microsoft.Resources/deployments/sentinel-lab-resources",
    "resourceGroup": "rg-sentinel-homelab",
    "resourceName": "sentinel-lab-resources",
  }
}

```

Wait for the deployment to finish. Record the output workspace name:

```

az deployment sub show \
  --name sentinel-homelab \
  --query properties.outputs \
  --output json

```

```

muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az deployment sub show \
> --name sentinel-homelab \
> --query properties.outputs \
> --output json
{
  "resourceGroupId": {
    "type": "String",
    "value": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab"
  },
  "workspaceName": {
    "type": "String",
    "value": "law-sentinel-homelab-g32ur33aju6wk"
  },
  "workspaceResourceId": {
    "type": "String",
    "value": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab/providers/Microsoft.OperationalInsights/workspaces/law-sentinel-homelab-g32ur33aju6wk"
  }
}
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$

```

Verify the resource group and resources:

```

az group show --name rg-sentinel-homelab --output table
az resource list --resource-group rg-sentinel-homelab --output
t table

```

```

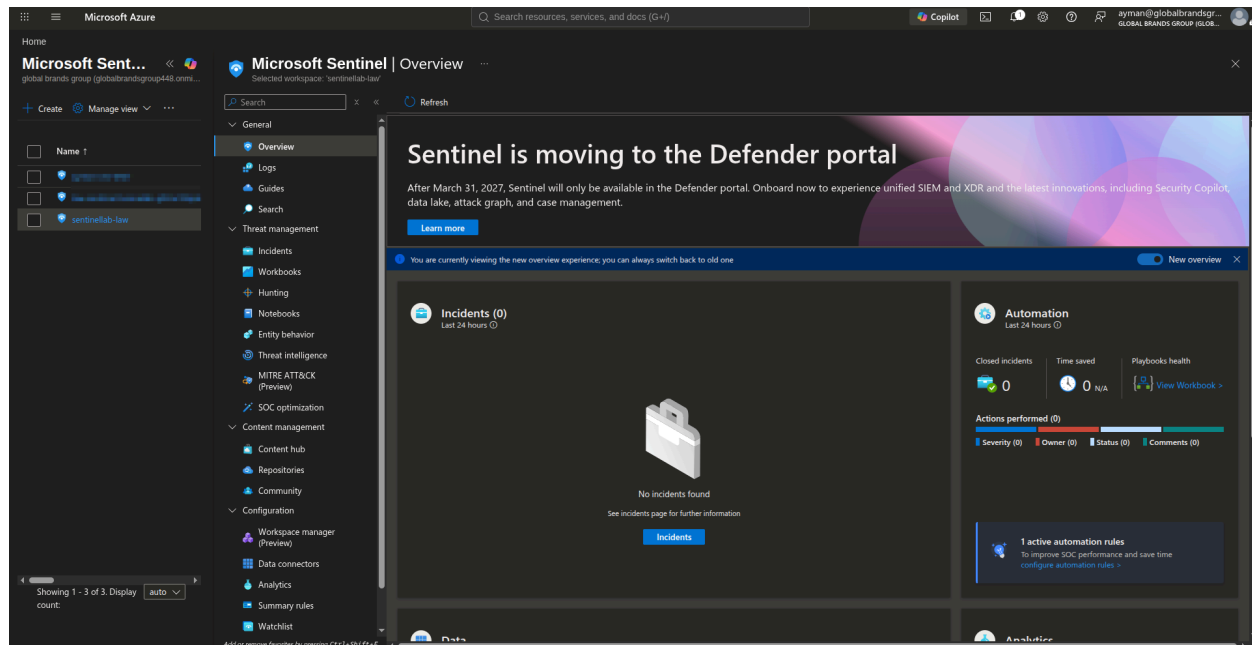
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az group show --name rg-sentinel-homelab --output table
Location      Name
-----
eastus        rg-sentinel-homelab
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az resource list --resource-group rg-sentinel-homelab --output table
Name
-----
law-sentinel-homelab-g32ur33aju6wk
SecurityInsights(law-sentinel-homelab-g32ur33aju6wk)
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$

```

Name	ResourceGroup	Location	Type	Status
law-sentinel-homelab-g32ur33aju6wk	rg-sentinel-homelab	eastus	Microsoft.OperationalInsights/workspaces	Succeeded
SecurityInsights(law-sentinel-homelab-g32ur33aju6wk)	rg-sentinel-homelab	eastus	Microsoft.OperationsManagement/solutions	Succeeded

9. Check Sentinel in the Azure Portal

1. Open the Azure portal.
2. Search for **Microsoft Sentinel**.
3. Open the workspace named by the deployment output.
4. Confirm that the workspace is onboarded.
5. Open **Content hub** and install Azure Activity.
6. Open Data Connector and configure Azure Activity.
7. Open **Analytics** and confirm the two rules beginning with **Lab -** are enabled.



Microsoft Defender

Search

Content hub

496 Solutions | 368 Standalone contents | 0 Installed | 0 Updates

Didn't find what you were looking for? We're showing a limited set of results. Try refining your search for more specific results. [Learn more](#)

Azure Activity

Status: All | Content type: All | Support: All | Provider: All | Category: All | Content sources: All

Install/Update | Delete

	Content title	Status	Content source	Provider	Support
☐	Azure Activity	Not installed	Solution	Microsoft	Microsoft
☐	Azure Activity	Not installed	Solution	Microsoft	Microsoft
☐	AzureActivity Administration from VPS Providers	Not installed	Solution	Microsoft	Microsoft
☐	Rare subscription-level operations in Azure	Not installed	Solution	Microsoft	Microsoft
☐	Azure storage key enumeration	Not installed	Solution	Microsoft	Microsoft
☐	Port opened for an Azure Resource	Not installed	Solution	Microsoft	Microsoft
☐	Anomalous Azure Operation Hunting Model	Not installed	Solution	Microsoft	Microsoft
☐	Azure Virtual Network Subnets Administrative ...	Not installed	Solution	Microsoft	Microsoft

Previous Page 1 of 2 Next > Showing 1 to 20 of 21 results.

Azure Activity

Microsoft Provider | Microsoft Support | 3.0.4 Version

Description

Note: Please refer to the following before installing the solution:

- Review the solution [Release Notes](#)

The Azure Activity solution for Microsoft Sentinel enables you to ingest Azure Activity Administrative, Security, Service Health, Alert Recommendation, Policy, Autoscale and Resource Health logs using Diagnostic Settings into Microsoft Sentinel.

Data Connectors: 1 Workbooks: 2, Analytic Rules: 14, Hunting Queries: 15

Learn more about Microsoft Sentinel | Learn more about Solutions

Content type

14 Analysis rule | 1 Data connector | 15 Hunting query

2 Workbook

Category

IT Operations

Pricing

Free

Install View details

Microsoft Defender

Search

My connectors - preview

Data Types: All | Status: All

Status	Connector name	Updates
	Azure Activity	...
	Microsoft 365 Insider ...	...

Disconnected Status | 4 Microsoft Provider | -- Last Log Received

Description

Azure Activity Log is a subscription log that provides insight into subscription-level events that occur in Azure, including events from Azure Resource Manager operational data, service health events, write operations taken on the resources in your subscription, and the status of activities performed in Azure.

Last data received

Content source

Azure Activity

Version

2.0.0

Author

Microsoft

Supported by

Microsoft Corporation | Email

Related content

0 Workbooks | 2 Queries | 15 Analytics rules templates

Data received

August 4

AzureActivity : 0

0 AzureActivity

Data types

AzureActivity --

Open connector page

Microsoft Defender

Search

Find Defender solutions

Home

Cases

Exposure management

Investigation & response

Threat intelligence

Assets

Microsoft Sentinel

Search

Threat management

Content management

Content hub

Repositories

Community

Configuration

Tables

Data connectors

Analytics

Summary rules

Watchlist

Automation

Identities

Endpoints

Email & collaboration

Cloud apps

Cloud security

STATUS

PROVIDER

Last Log received

Description

Azure Activity Log is a subscription log that provides insight into subscription-level events that occur in Azure, including events from Azure Resource Manager operational data, service health events, write operations taken on the resources in your subscription, and the status of activities performed in Azure.

Last data received

Content source

Azure Activity

Version

2.0.0

Author

Microsoft

Supported by

Microsoft Corporation | Email

Related content

0 Workbooks

2 Queries

15 Analytics rules templates

Data received

Go to log analytics

0

AzureActivity

Data types

AzureActivity --

To integrate with Azure Activity make sure you have:

- ✓ **Workspace:** read and write permissions.
- ① **Policy:** owner role assigned for each policy assignment scope.
- ② **Subscription:** owner role permission on the relevant subscription

Table management

Manage the table tiers - analytics lake and table retention. Select the table to manage the table tier and configure its retention settings.

0 items Refresh

Table name Tier Table type Analytics reten... Total retention Workspace

No data available

Configuration

This connector has been updated to use the diagnostics settings back-end pipeline, which provides increased functionality and better consistency with resource logs. Connectors using this pipeline can also be governed at scale by Azure Policy. Learn more about the new Azure Activity connector. Follow the instructions below to upgrade your connector to the diagnostics settings pipeline.

2. Connect your subscriptions through diagnostic settings new pipeline

This connector uses Azure Policy to apply a single Azure Subscription log-streaming configuration to a collection of subscriptions, defined as a scope. Follow the instructions below to create and apply a policy to all current and future subscriptions. **Note**, you may already have an active policy for this resource type.

Launch the Azure Policy Assignment wizard and follow the steps.

1. In the **Basics** tab, click the button with the three dots under **Scope** to select your resources assignment scope. >2. In the **Parameters** tab, choose your Microsoft Sentinel workspace from the **Log Analytics workspace** drop-down list, and leave marked as "true" all the log and metric types you want to ingest. >3. To apply the policy on your existing resources, select the **Remediation** tab and mark the **Create a remediation task** checkbox.

Launch Azure Policy Assignment wizard

Microsoft Azure

Search resources, services, and docs (G+)

Home

Assign policy

Basics Parameters Remediation Managed identity Non-compliance messages Review + create

Scope

Scope *

Learn more about setting the scope

Exclusions

Optionally select resources to exclude from the policy assignment.

Resource selectors

Using resource selectors, you can further refine this assignment's applicability by targeting specific subsets of resources. Expand to learn more.

Basics

Policy definition *

Configure Azure Activity logs to stream to specified Log Analytics w...

Version *

Overrides

Using overrides, you can change the effects or referenced versions of definitions for all or a subset of resources evaluated by this assignment. Expand to learn more.

Assignment name *

Configure Azure Activity logs to stream to specified Log Analytics worksp...

Description

Policy enforcement

Default

Do not enforce

Enroll

Scope

Subscription

Ayman-Visual Studio Enterprise Subscription - MPN

Resource Group

rg-sentinel-homelab

Select Cancel Clear All Selections Give feedback

Microsoft Azure

Search resources, services, and docs (G+)

Home

Assign policy

Basics Parameters Remediation Managed identity Non-compliance messages Review + create

Scope

Scope * Ayman-Visual Studio Enterprise Subscription - MPN/rg-sentinel-ho... [Learn more about setting the scope](#)

Exclusions [Optionally select resources to exclude from the policy assignment](#)

Resource selectors [Expand](#) Using resource selectors, you can further refine this assignment's applicability by targeting specific subsets of resources. Expand to learn more.

Basics

Policy definition * [Configure Azure Activity logs to stream to specified Log Analytics w...](#)

Version * [1.0.0](#)

Overrides [Expand](#) Using overrides, you can change the effects or referenced versions of definitions for all or a subset of resources evaluated by this assignment. Expand to learn more.

Assignment name * [Configure Azure Activity logs to stream to specified Log Analytics worksp...](#)

Description

Policy enforcement [Default](#)
[Do not enforce](#)
[Enroll](#)

Previous Next **Review + create**

Select policy version

This policy definition has the following versions. Select the one you would like to assign. Learn more about policy definition versions [Learn more](#)

The option you select will determine the major version this assignment will reference. You can choose whether to automatically enroll in minor version upgrades and/or previews. All assignments will always receive patch changes.

Version updates

Automatically enroll in minor version changes [No](#)

Version 1	Assignments 1
1.0.0	0

Include preview versions [No](#)

Selected version [1.0.0](#)

Select Cancel [Give feedback](#)

Microsoft Azure

Search resources, services, and docs (G+)

Home

Assign policy

Basics Parameters Remediation Managed identity Non-compliance messages Review + create

Scope

Scope * Ayman-Visual Studio Enterprise Subscription - MPN/rg-sentinel-ho... [Learn more about setting the scope](#)

Exclusions [Optionally select resources to exclude from the policy assignment](#)

Resource selectors [Expand](#) Using resource selectors, you can further refine this assignment's applicability by targeting specific subsets of resources. Expand to learn more.

Basics

Policy definition * [Configure Azure Activity logs to stream to specified Log Analytics w...](#)

Version * [1.0.0](#)

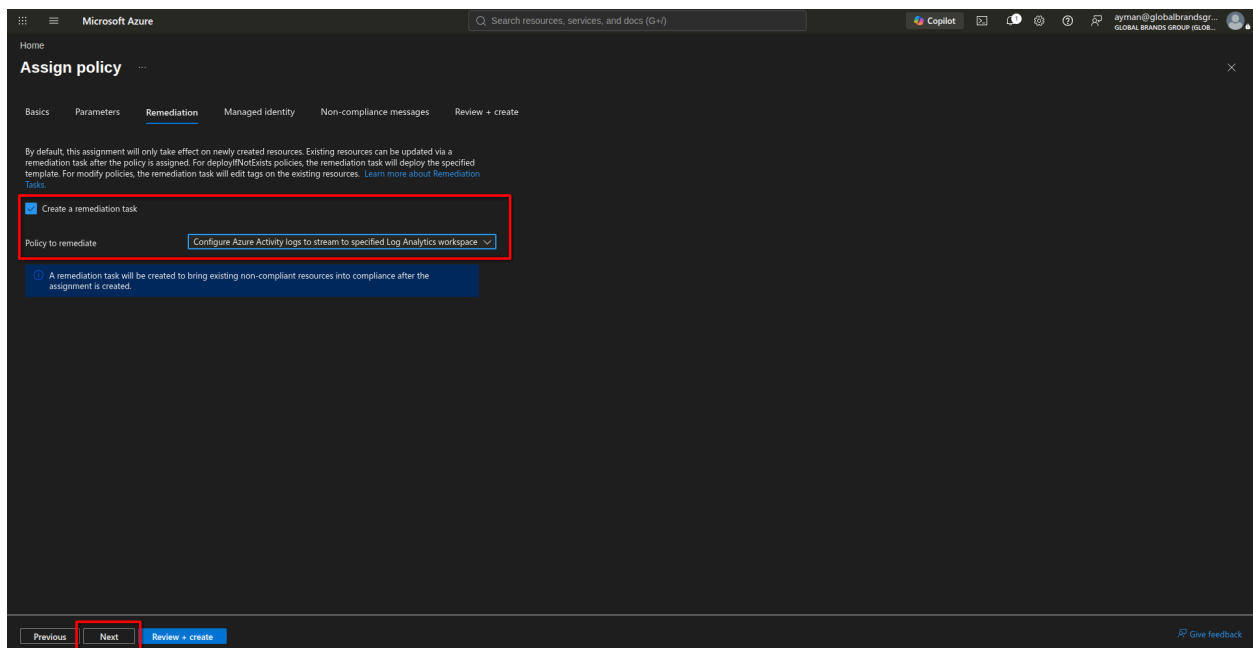
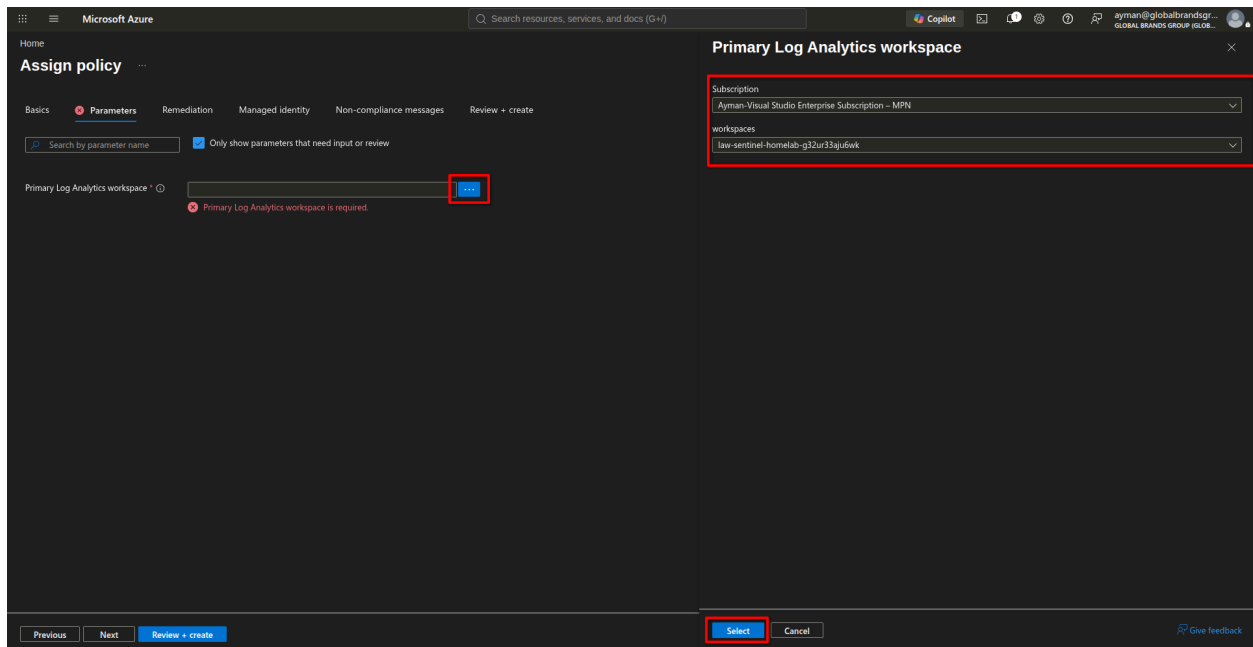
Overrides [Expand](#) Using overrides, you can change the effects or referenced versions of definitions for all or a subset of resources evaluated by this assignment. Expand to learn more.

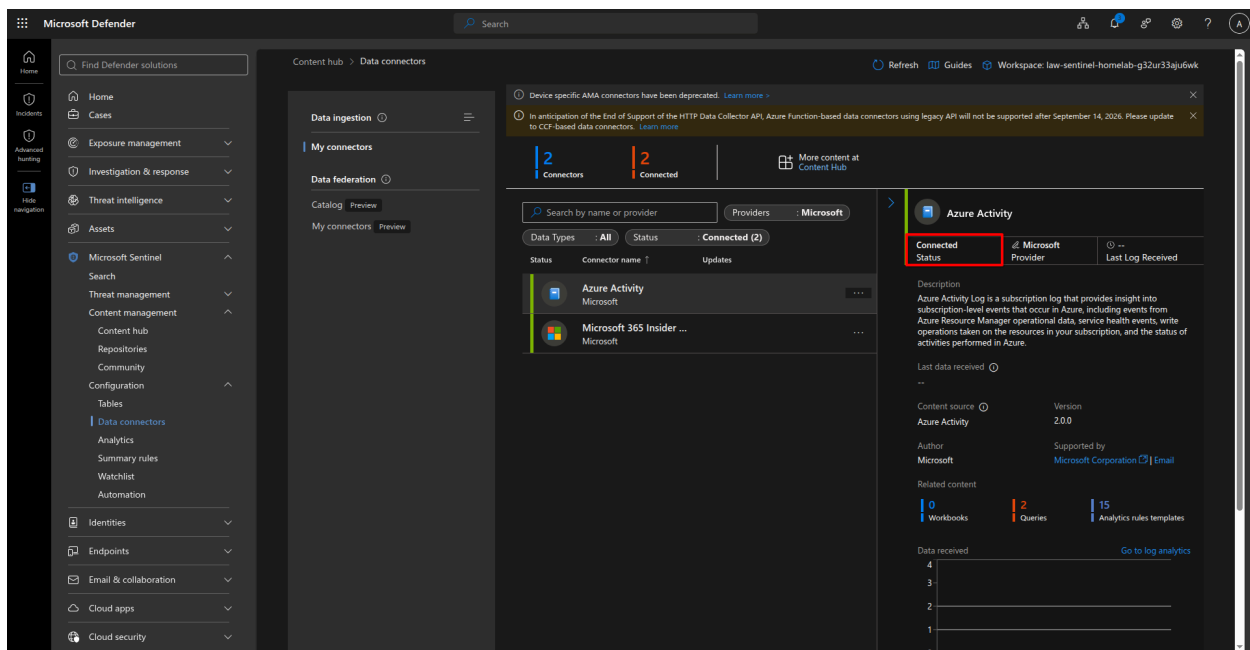
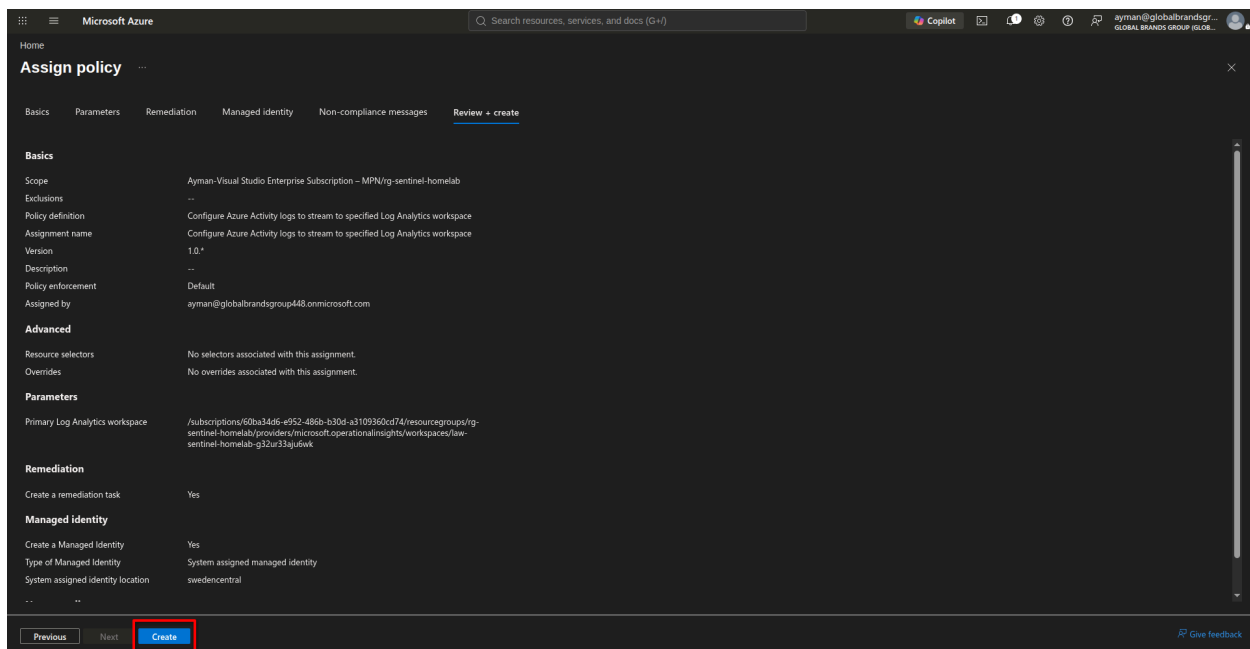
Assignment name * [Configure Azure Activity logs to stream to specified Log Analytics worksp...](#)

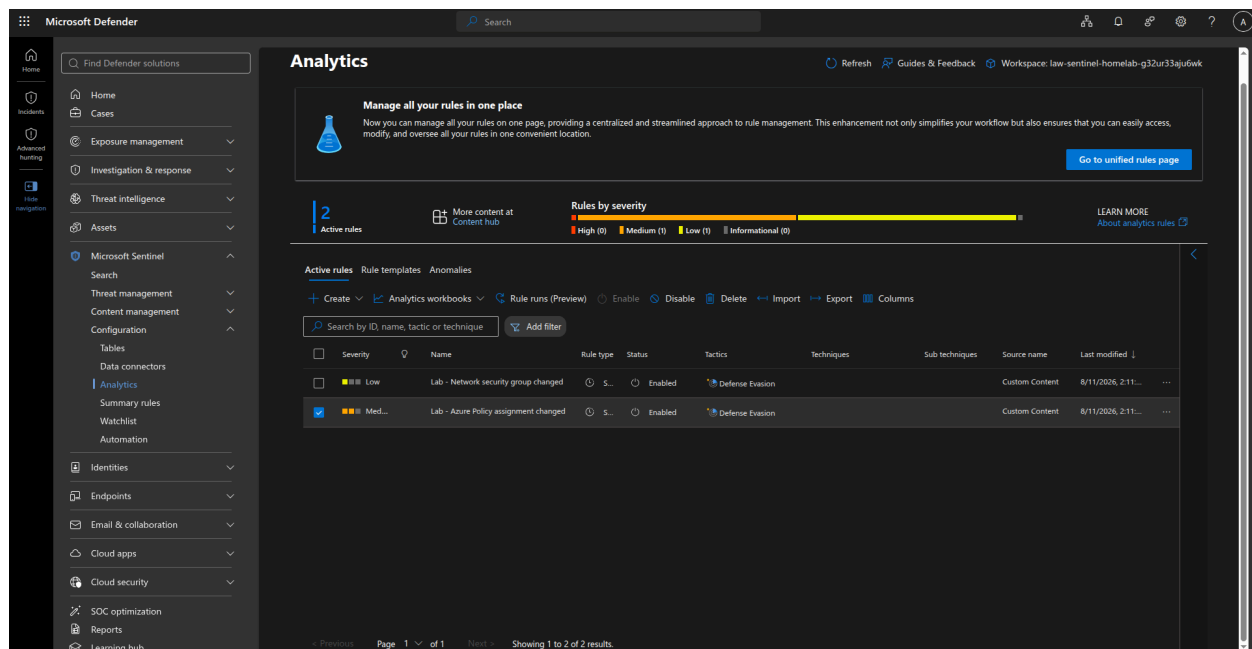
Description

Policy enforcement [Default](#)
[Do not enforce](#)
[Enroll](#)

Previous **Next** [Review + create](#) [Give feedback](#)



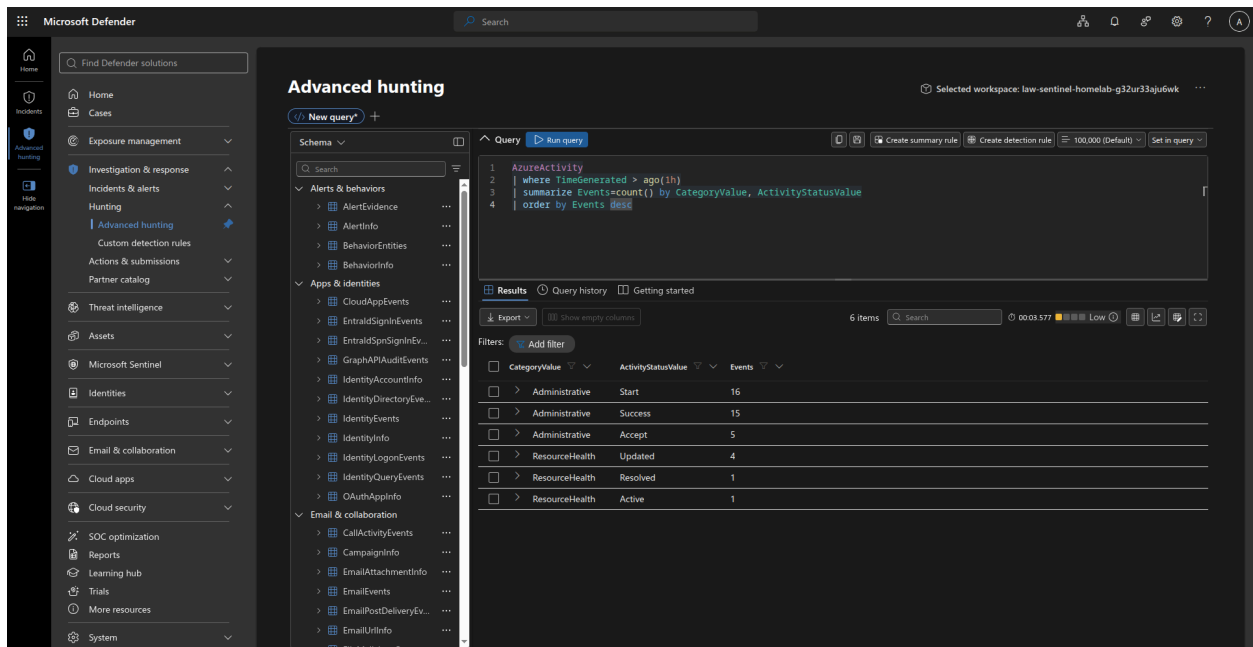




10. Verify Activity Log Ingestion with KQL

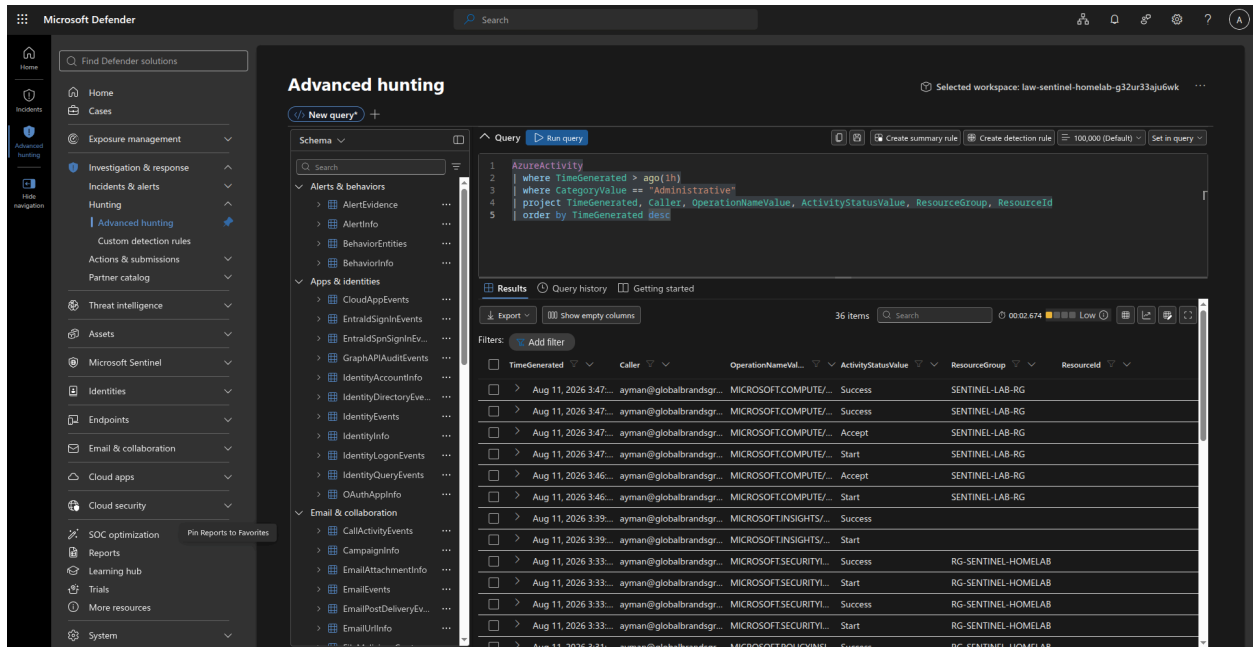
In the Sentinel workspace, open **Logs** and run:

```
AzureActivity
| where TimeGenerated > ago(1h)
| summarize Events=count() by CategoryValue, ActivityStatusValue
| order by Events desc
```



Show recent administrative events:

```
AzureActivity
| where TimeGenerated > ago(1h)
| where CategoryValue == "Administrative"
| project TimeGenerated, Caller, OperationNameValue, ActivityStatusValue, ResourceGroup, ResourceId
| order by TimeGenerated desc
```



If there are no rows, confirm that the subscription diagnostic setting exists and wait up to 15 minutes before testing again.

11. Generate a Safe Test Event

Use a separate temporary resource group. This avoids changing an existing application resource.

Create the test resource group:

```
az group create \
  --name rg-sentinel-test-events \
  --location eastus \
  --tags environment=homelab purpose=sentinel-test
```

```

muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az group create \
> --name rg-sentinel-test-events \
> --location eastus \
> --tags environment=homelab purpose=sentinel-test
{
  "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-test-events",
  "location": "eastus",
  "managedBy": null,
  "name": "rg-sentinel-test-events",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": {
    "environment": "homelab",
    "purpose": "sentinel-test"
  },
  "type": "Microsoft.Resources/resourceGroups"
}
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$

```

Create a temporary network security group:

```

az network nsg create \
  --resource-group rg-sentinel-test-events \
  --name nsg-sentinel-test

```

```

muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az network nsg create \
> --resource-group rg-sentinel-test-events \
> --name nsg-sentinel-test
{
  "NewNSG": {
    "defaultSecurityRules": [
      {
        "access": "Allow",
        "description": "Allow inbound traffic from all VMs in VNET",
        "destinationAddressPrefix": "VirtualNetwork",
        "destinationAddressPrefixes": [],
        "destinationPortRange": "**",
        "destinationPortRanges": [],
        "direction": "Inbound",
        "etag": "W/\"bb7d309f-a01a-434f-8b63-abb6837e11e\"",
        "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-test-events/providers/Microsoft.Network/networkSecurityGroups/nsg-sentinel-test/defaultSecurityRules/AllowVnetInBound",
        "name": "AllowVnetInBound",
        "priority": 65000,
        "protocol": "**",
        "provisioningState": "Succeeded",
        "resourceGroup": "rg-sentinel-test-events",
        "sourceAddressPrefix": "VirtualNetwork",
        "sourceAddressPrefixes": [],
        "sourcePortRange": "**",
        "sourcePortRanges": [],
        "type": "Microsoft.Network/networkSecurityGroups/defaultSecurityRules"
      },
      {
        "access": "Allow",
        "description": "Allow inbound traffic from azure load balancer",
        "destinationAddressPrefix": "**",
        "destinationAddressPrefixes": [],
        "destinationPortRange": "**",
        "destinationPortRanges": [],
        "direction": "Inbound",
        "etag": "W/\"bb7d309f-a01a-434f-8b63-abb6837e11e\"",
        "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-test-events/providers/Microsoft.Network/networkSecurityGroups/nsg-sentinel-test/defaultSecurityRules/AllowAzureLoadBalancerInBound",
        "name": "AllowAzureLoadBalancerInBound",
        "priority": 65001,
        "protocol": "**",
        "provisioningState": "Succeeded",
        "resourceGroup": "rg-sentinel-test-events",

```

Update it with a harmless description:


```
az network nsg update \
  --resource-group rg-sentinel-test-events \
  --name nsg-sentinel-test \
  --tags testChange=updated
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az network nsg update \
> --resource-group rg-sentinel-test-events \
> --name nsg-sentinel-test \
> --tags testChange=updated
{
  "defaultSecurityRules": [
    {
      "access": "Allow",
      "description": "Allow inbound traffic from all VMs in VNET",
      "destinationAddressPrefix": "VirtualNetwork",
      "destinationAddressPrefixes": [],
      "destinationPortRange": "**",
      "destinationPortRanges": [],
      "direction": "Inbound",
      "etag": "W/\"7b17692a-4cc2-4e03-a7a6-238bd7456ce1\"",
      "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-test-events/providers/Microsoft.Network/networkSecurityGroups/nsg-sentinel-test/defaultSecurityRules/AllowVnetInBound",
      "name": "AllowVnetInBound",
      "priority": 65000,
      "protocol": "**",
      "provisioningState": "Succeeded",
      "resourceGroup": "rg-sentinel-test-events",
      "sourceAddressPrefix": "VirtualNetwork",
      "sourceAddressPrefixes": [],
      "sourcePortRange": "**",
      "sourcePortRanges": [],
      "type": "Microsoft.Network/networkSecurityGroups/defaultSecurityRules"
    },
    {
      "access": "Allow",
      "description": "Allow inbound traffic from azure load balancer",
      "destinationAddressPrefix": "**",
      "destinationAddressPrefixes": [],
      "destinationPortRange": "**",
      "destinationPortRanges": [],
      "direction": "Inbound",
      "etag": "W/\"7b17692a-4cc2-4e03-a7a6-238bd7456ce1\"",
      "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-test-events/providers/Microsoft.Network/networkSecurityGroups/nsg-sentinel-test/defaultSecurityRules/AllowAzureLoadBalancerInBound",
      "name": "AllowAzureLoadBalancerInBound",
      "priority": 65001,
      "protocol": "**",
      "provisioningState": "Succeeded",
      "resourceGroup": "rg-sentinel-test-events",

```

Delete the test network security group:

```
az network nsg delete \
  --resource-group rg-sentinel-test-events \
  --name nsg-sentinel-test
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az network nsg delete \
> --resource-group rg-sentinel-test-events \
> --name nsg-sentinel-test
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$
```

Delete the temporary group after the events have been generated:

```
az group delete \
  --name rg-sentinel-test-events \
```

```
--yes \  
--no-wait
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az group delete \  
> --name rg-sentinel-test-events \  
> --yes \  
> --no-wait  
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$
```

Do not create public-facing machines, malicious files, brute-force traffic, or exploit traffic for this starter lab.

12. Confirm the Network Detection

Wait several minutes, then run:

```
AzureActivity  
| where TimeGenerated > ago(30m)  
| where ResourceProviderValue =~ "Microsoft.Network"  
| where OperationNameValue has_any ("networkSecurityGroups", "securityGroups")  
| project TimeGenerated, Caller, OperationNameValue, ActivityStatusValue, ResourceGroup,  
ResourceId  
| order by TimeGenerated desc
```

The screenshot displays the Microsoft Defender Advanced Hunting interface. On the left is a navigation pane with categories like Home, Incidents, Exposure management, Investigation & response, Threat intelligence, Assets, Microsoft Sentinel, Identities, Endpoints, Email & collaboration, Cloud apps, Cloud security, SOC optimization, Reports, Learning hub, Trials, and More resources. The main area is titled 'Advanced hunting' and shows a query editor with the following query:

```
1 AzureActivity  
2 | where TimeGenerated > ago(30m)  
3 | where ResourceProviderValue =~ "Microsoft.Network"  
4 | where OperationNameValue has_any ("networkSecurityGroups", "securityGroups")  
5 | project TimeGenerated, Caller, OperationNameValue, ActivityStatusValue, ResourceGroup, ResourceId  
6 | order by TimeGenerated desc
```

Below the query editor, the 'Results' tab shows 2 items. The results table has columns for TimeGenerated, Caller, OperationNameValue, ActivityStatusValue, ResourceGroup, and ResourceId. The results are as follows:

TimeGenerated	Caller	OperationNameValue	ActivityStatusValue	ResourceGroup	ResourceId
Aug 11, 2026 3:59...	ayman@globalbrandsgr...	MICROSOFTNETWORK/...	Accept	RG-SENTINEL-TEST-EVE...	
Aug 11, 2026 3:59...	ayman@globalbrandsgr...	MICROSOFTNETWORK/...	Start	RG-SENTINEL-TEST-EVE...	

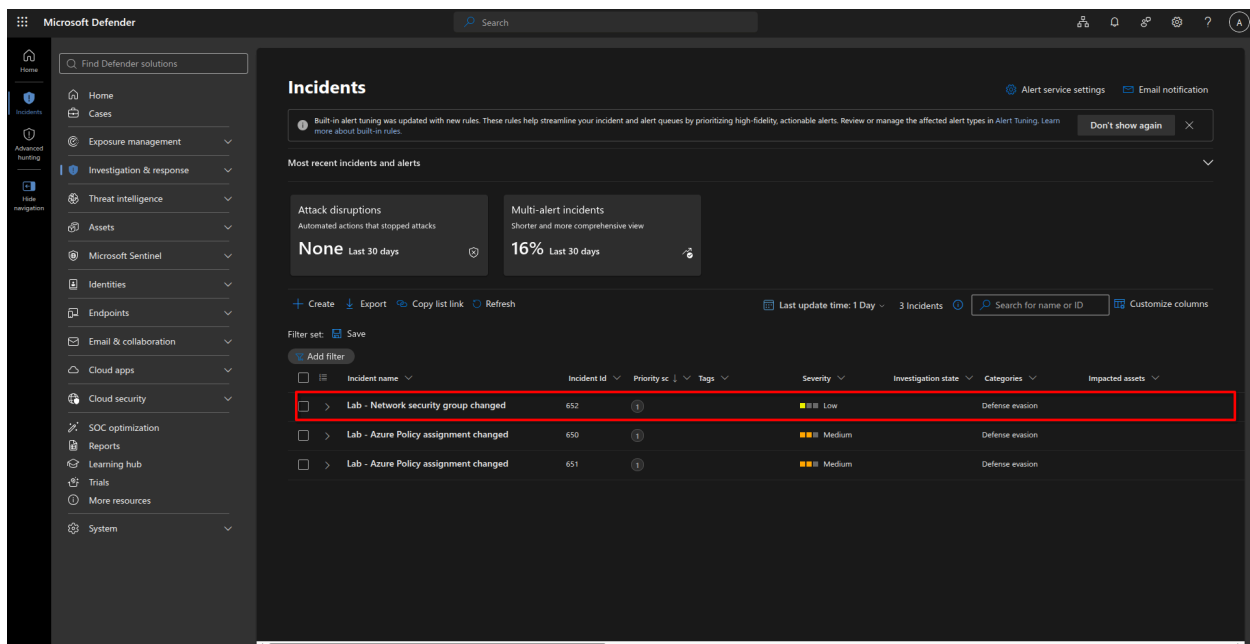
Then open **Microsoft Sentinel > Incidents** and look for:

Lab - Network security group changed

Open the incident and inspect the alert, entities, timeline, caller, operation name, and affected resource.

If the Activity Log query returns an event but no incident appears, check that the analytics rule is enabled and that its lookback window includes the event.

Scheduled rules can take another evaluation cycle to create an incident.



The screenshot displays the Microsoft Sentinel 'Incidents' page. The left sidebar contains navigation options like Home, Cases, Exposure management, Investigation & response, Threat intelligence, Assets, Microsoft Sentinel, Identities, Endpoints, Email & collaboration, Cloud apps, Cloud security, SOC optimization, Reports, Learning hub, Trials, More resources, and System. The main content area shows a summary of incidents and a table of recent incidents.

Incidents

Built-in alert tuning was updated with new rules. These rules help streamline your incident and alert queues by prioritizing high-fidelity, actionable alerts. Review or manage the affected alert types in Alert Tuning. [Learn more about built-in rules.](#) [Don't show again](#)

Most recent incidents and alerts

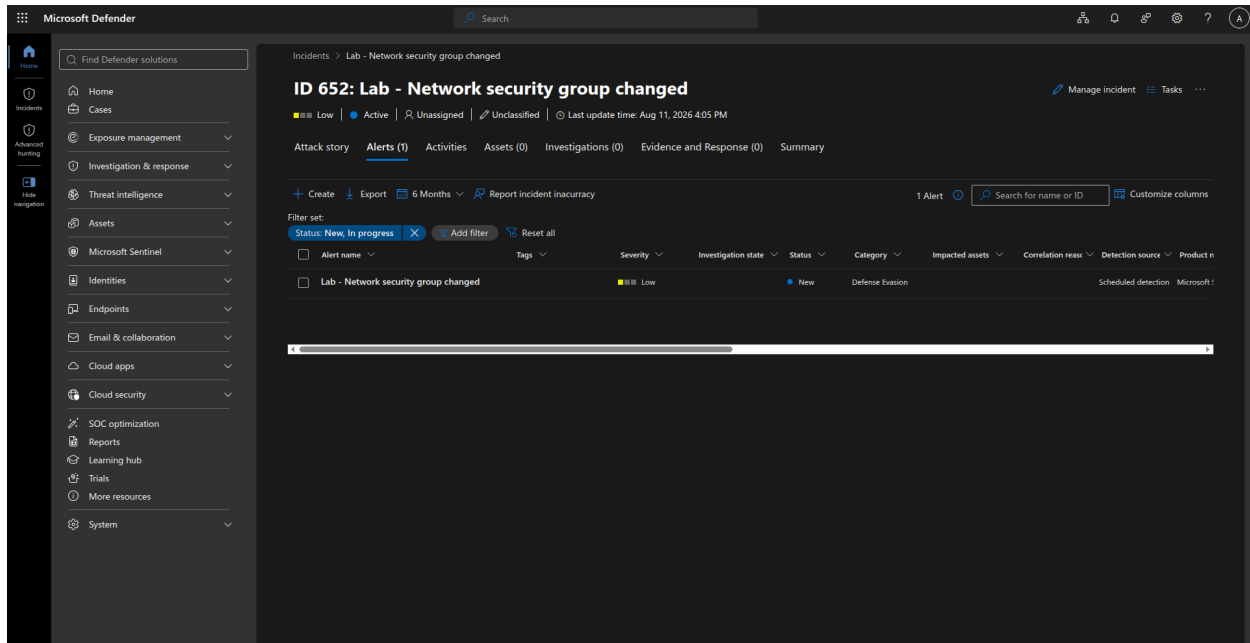
Attack disruptions
Automated actions that stopped attacks
None Last 30 days

Multi-alert incidents
Shorter and more comprehensive view
16% Last 30 days

+ Create | Export | Copy list link | Refresh | Last update time: 1 Day | 3 Incidents | Search for name or ID | Customize columns

Filter set: Save | Add Filter

Incident name	Incident Id	Priority	Severity	Investigation state	Categories	Impacted assets
Lab - Network security group changed	652	1	Low		Defense evasion	
Lab - Azure Policy assignment changed	650	1	Medium		Defense evasion	
Lab - Azure Policy assignment changed	651	1	Medium		Defense evasion	



13. Clean Up the Lab

First remove the temporary test group if it still exists:

```
az group delete \
  --name rg-sentinel-test-events \
  --yes \
  --no-wait
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az group delete \
> --name rg-sentinel-test-events \
> --yes \
> --no-wait
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$
```

Delete the subscription-level Activity Log diagnostic setting. The Bicep deployment is not being used here to delete resources, so remove this explicitly:

```
az monitor diagnostic-settings subscription delete \
  --name send-activity-to-sentinel
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az monitor diagnostic-settings subscription delete \
--name send-activity-to-sentinel
Are you sure you want to perform this operation? (y/n): y
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$
```

Delete the lab resource group:

```
az group delete \
  --name rg-sentinel-homelab \
  --yes \
  --no-wait
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az group delete \
  --name rg-sentinel-homelab \
  --yes \
  --no-wait
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$
```

14. Troubleshooting

Bicep validation fails

- Update Azure CLI and Bicep.
- Confirm the selected region supports Log Analytics and Microsoft Sentinel.
- Confirm your account has the required permissions.
- Check that resource names are unique where required.

Sentinel is not visible

- Confirm the workspace resource exists.
- Confirm the `onboardingStates/default` resource deployed successfully.
- Open Microsoft Sentinel from the workspace rather than searching only by resource group.
- Check the deployment operation details for authorization errors.

AzureActivity is empty

- Confirm the subscription diagnostic setting points to the deployed workspace.
- Confirm the enabled categories include `Administrative`.
- Wait 5 to 15 minutes after the first deployment.
- Generate a new test event after the diagnostic setting is deployed.

The query returns events but no incident

- Confirm the rule is enabled under **Analytics**.
- Check the rule's query frequency and lookback period.
- Verify the event matches the query exactly.
- Wait for at least one scheduled evaluation cycle.
- Check **Incidents** and **Alerts**, not only the Logs result.

Deployment creates an authorization error

- Ask the subscription administrator which role is missing.
- Remember that subscription diagnostic settings and Sentinel permissions may be separate from resource-group Contributor access.
- Do not grant permanent Owner access solely to complete a lab if a narrower role is available.

15. Next Exercises

After the starter lab works, extend it one feature at a time:

- Add a Microsoft Entra ID connector and test sign-in detections.
- Add a storage account and enable only the logs needed for a specific exercise.
- Add a small Linux VM with automatic shutdown, then monitor SSH authentication logs.
- Create a Logic App playbook with a managed identity for incident notification.
- Move analytics rules into separate Bicep modules.
- Add Azure Policy requiring `environment` and `managedBy` tags.

- Store the Bicep project in Git and deploy it from GitHub Actions or Azure DevOps.
- Add a CI pipeline that runs `az bicep build` and subscription deployment validation on every pull request.